

candidate failure mode	measurement needed before model change
Low-T2 open-pore removal too slow	interrupted D50/D90, connectivity and shrinkage rate
Low-T2 closed-pore shrinkage too slow	open/closed pore fractions and gas pressure at switch
Low-T2 nucleation/activity too low	low-T creep/nucleation-rate data under relevant stress
Surface coarsening/PR too strong before or during step 2	interrupted pore-size distributions through both steps
Fine-pore pinning too weak	simultaneous grain and pore distributions at switch/hold
Zener pinning released too early	boundary mobility and pore-boundary attachment data
Clean grain-growth mobility too high	pore-free/low-porosity grain-growth kinetics
Initial common pore scale too coarse	950 C D50/D90 and connectivity
Closed-pore fraction/accommodation missing or uncalibrated	tomography/dilatometry identifying closure and trapped gas
Barrier extrapolation suppresses low-T2 activity	barrier/creep data at 1000–1300 C and relevant stress